

Requirement Analysis Phase

Solution Requirements: Functional & Non-Functional Specifications

Attribute	Project Information
Project Title	Auto Ticket Classification using Flow Designer
Phase / Stage	2. Requirement Analysis
Document Title	Solution Requirements (Functional & Non-Functional)
Platform & Scope	ServiceNow Utah / Vancouver / Washington DC (Global Scope)
Team ID	SWTID-2026-4988

1. Functional Requirements (FR)

The proposed ServiceNow solution must fulfill the following explicit functional capabilities:

FR No.	Functional Requirement (Epic)	Sub-Requirement / Detailed Specification	Target Field / Component	Verification Acceptance Criteria
FR-1	Custom Data Architecture	Create a dedicated custom ServiceNow table labeled "Incident WorkFlow" with technical name "u_incident_workflow".	Table: u_incident_workflow Scope: Global	Table successfully created under System Definition > Tables and visible in application navigator.
FR-2	Automated Record Numbering	Configure auto-numbering controls on the custom table with prefix "INC", starting number 500, and 5-digit padding.	Controls tab: Auto Number Prefix: INC Number: 500 Digits: 5	New records generate consecutive numbers: INC00509, INC00510, INC00511, INC00512 without collisions.
FR-3	Data Model & Field Definition	Create all operational fields: Caller (Reference: sys_user), Short Description (String), Description (String), Category (Choice), Subcategory (Choice), State (Choice), Assigned Group (Reference: sys_user_group), Assigned To (Reference: sys_user).	Form Designer / Dictionary Entries	All fields display with appropriate controls; Caller and Short Description are marked mandatory.

FR No.	Functional Requirement (Epic)	Sub-Requirement / Detailed Specification	Target Field / Component	Verification Acceptance Criteria
FR-4	Dynamic Dependent Choice Logic	Configure Subcategory choices to dynamically depend on Category: Network -> Wi-Fi; Hardware -> Projector; Access -> Forgot Password; Performance -> Slow Computer.	Dictionary Entry: u_subcategory Section: Dependent Field Dependent on: u_category	Selecting "Network" displays only "Wi-Fi"; selecting "Hardware" displays only "Projector"; selecting "Access" displays only "Forgot Password"; selecting "Performance" displays only "Slow Computer".
FR-5	Event-Driven Flow Trigger	Configure a Flow Designer flow titled "Auto Classify School IT Tickets" triggered when a record is created on Incident Workflow and Category is empty.	Trigger: Record Created Table: u_incident_workflow Condition: Category is Empty	Flow triggers automatically upon record insertion if and only if Category is unpopulated.
FR-6	Keyword Classification Logic	Implement conditional branches in Flow Designer evaluating keywords in Short Description: "Wi-Fi"/"Network" -> Network/Wi-Fi; "Projector" -> Hardware/Projector; "Forgot password" -> Access/Forgot Password; "Slow computer" -> Performance/Slow Computer.	Flow Logic: If / Else If Action: Update Record	When short description contains target keyword, record fields Category and Subcategory are updated correctly.
FR-7	Automated Caller Notification	Configure Flow Designer Action "Send Email" sending confirmation email to Caller with subject "Your Request for the issue has been submitted."	Action: Send Email To: Caller -> Email Subject & Body dynamic	Email record appears in System Logs > Emails with recipient email and previewable confirmation body.
FR-8	Application Lifecycle Packaging	Capture all configurations in a dedicated ServiceNow Update Set labeled "Project Update Set" and export to XML upon completion.	System Update Sets > Local Update Sets State: Complete > Export to XML	Generates valid XML update set file containing all table, dictionary, form, and flow artifacts.

2. Non-Functional Requirements (NFR)

The non-functional specifications define the system's quality attributes, operational limits, security posture, and usability benchmarks:

NFR No.	Non-Functional Dimension	Specification & Benchmark Requirement	Enforcement Mechanism
NFR-1	Performance Execution Speed &	Flow Designer execution latency must be under 1.5 seconds from record insertion to completion of categorization and email dispatch.	ServiceNow asynchronous execution engine; lightweight no-code Flow Studio logic.
NFR-2	Reliability Determinism &	Classification logic must execute with 100% deterministic accuracy for specified keywords with zero false-positive field overwrites.	Exact condition matching in Flow Designer and defensive checks (Category is Empty trigger condition).
NFR-3	Usability Accessibility &	End-user incident submission form must require no more than two inputs (Caller and Short Description) with zero technical jargon required.	Simplified Form Designer layout hiding internal triage parameters from initial caller view.
NFR-4	Security & Role-Based Access	Access to custom Incident Workflow records governed by role u incident workflow user; read/write access restricted according to standard ACLs.	ServiceNow Access Control Lists (ACLs) generated automatically on custom table creation.
NFR-5	Maintainability & Extensibility	100% no-code architecture allowing administrators to add new keywords, categories, or email templates without writing JavaScript.	ServiceNow Flow Designer natural language building blocks and Choice List Dictionary configuration.
NFR-6	Auditability & Logging	All classification decisions, execution runs, and outbound email deliveries must be comprehensively logged and queryable for audits.	ServiceNow Flow Designer Execution History and System Logs > Emails (sys_email) repository.